Chapter 11: Derivatives of Multivariable Functions (§11.5 - 11.8)

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Contents

So far, we have interpreted the double integral of a function f over a domain D in two different ways.

- First, $\iint_D f(x,y)dA$ tells us a difference of volumes - the volume the surface defined by f bounds above the xy-plane on D minus the volume the surface bounds below the xy-plane on D.
- Additionally, $\frac{1}{A(D)}\iint_D f(x,y)dA$ determines the average value of d on D.

Preview activity

Suppose that we have a flat, thin object (called a **lamina**) whose density varies across the object. We can think of the density on a lamina as a measure of mass per unit area. As an example, consider a circular plate D of radius 1 cm centered at the origin whose density δ varies depending on the distance from its center so that the density in grams per square centimeter at point (x, y) is

$$\delta(x, y) = 10 - 2(x^2 + y^2).$$

a. Suppose that we partition the plate into subrectangles R_{ij}, where $1 \leq i \leq m$ and $1 \leq j \leq n$, of equal area ΔA, and select a point (x_{ij}^*, y_{ij}^*) in R_{ij} for each i and j. What is the meaning of the quantity $\delta(x_{ij}^*, y_{ij}^*)\Delta A$?

b. State a double Riemann sum that provides an approximation of the mass of the plate.

c. Explain why the double integral

$$\iint_D \delta(x,y)\, dA$$

tells us the exact mass of the plate.

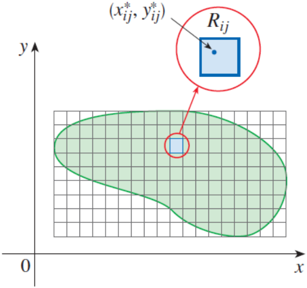
d. Determine an iterated integral which, if evaluated, would give the exact mass of the plate. Do not actually evaluate the integral. (This integral is considerably easier to evaluate in polar coordinates, which we will learn more about in **Section 11.5**.)

Density and Mass

Suppose the lamina occupies a region D of the xy-plane and its density (in units of mass per unit area) at a point (x, y) in D is given by $\delta(x, y)$, where δ is a continuous function on D. This means that

$$\delta(x, y) = \lim_{\Delta A \to 0} \frac{\Delta m}{\Delta A}.$$

where Δm and ΔA are the mass and area of a small rectangle that contains (x, y).



To find the total mass m of the lamina D, we divide a rectangle R containing D into subrectangles R_{ij} of the same size. The approximation of total mass is

$$m \approx \sum_i \sum_j \delta(x_{ij}^*, y_{ij}^*) \Delta A.$$

If we now increase the number of subrectangles, we obtain the total mass m of the lamina as the limiting value of the approximations:

$$\boxed{m = \iint_D \delta(x, y) dA}$$

Activity 11.4.2. Let D be a half-disk lamina of radius 3 in quadrants IV and I, centered at the origin as shown in Figure 11.4.1. Assume the density at point (x, y) is given by $\delta(x, y) = x$. Find the exact mass of the lamina.

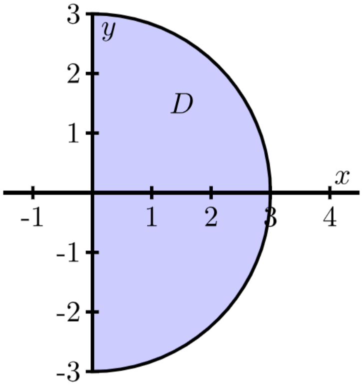


Figure 11.4.1. A half disk lamina.

Area

If we consider the situation where the mass-density distribution is constant, we can also see how a double integral may be used to determine the area of a region. Assuming that $\delta(x, y) = 1$ over a closed bounded region D, where the units of δ are "mass per unit of area," it follows that $\iint_D 1 dA$ is the mass of the lamina. But since the density is constant, the numerical value of the integral is simply the area.

Example

Suppose we want to find the area of the bounded region D between the curves

$$y = 1 - x^2 \quad \text{and} \quad y = x - 1.$$

a. The volume of a solid with constant height is given by the area of the base times the height. Hence, we may interpret the area of the region D as the volume of a solid with base D and of uniform height 1. That is, the area of D is given by $\iint_D 1\,dA$. Write an iterated integral whose value is $\iint_D 1\,dA$. (Hint: Which order of integration might be more efficient? Why?)

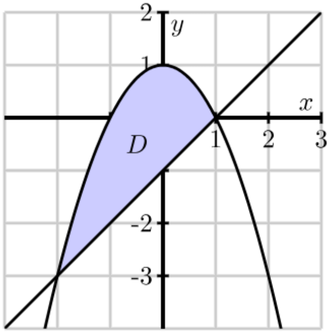


Figure 11.4.2. The graphs of $y = 1 - x^2$ and $y = x - 1$.

b. Evaluate the iterated integral from (a). What does the result tell you?

The double integral and area

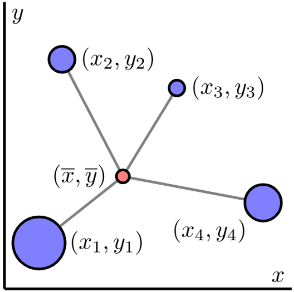
Given a closed, bounded region D in the plane, the area of D, denoted $A(D)$, is given by the double integral

$$A(D) = \iint_D 1 dA.$$

Center of Mass

The center of mass of an object is a point at which the object will balance perfectly. we start by finding the center of mass of a collection of N distinct point-masses in the plane.

Let $m_1, m_2, \ldots, m_N$ be N masses located in the plane. Think of these masses as connected by rigid rods of negligible weight from some central point (x, y).



Now imagine balancing this system by placing it on a thin pole at the point (x, y) perpendicular to the plane containing the masses. Unless the masses are perfectly balanced, the system will fall off the pole. The point $(\bar{x}, \bar{y})$ at which the system will balance perfectly is called the **center of mass** of the system. Our goal is to determine the center of mass of a system of discrete masses, then extend this to a continuous lamina.

- Each mass exerts a force (called a *moment*) around the lines $x = \bar{x}$ and $y = \bar{y}$ that causes the system to tilt in the direction of the mass.
- These moments are dependent on the mass and the distance from the given line.

Let (x_i, y_i) be the location of mass m_i, $i = 1, 2 \ldots, N$. In order to balance perfectly, the moments in the x direction and in the y direction must be in equilibrium. We determine these moments and solve the resulting system to find the equilibrium point $\bar{x}, \bar{y}$ at the center of mass.

- The force that mass m_1 exerts to tilt the system from the line $y = \bar{y}$ is $m_1 g(\bar{y} - y_1)$, where g is the gravitational constant.
- Similarly, the force that mass m_1 exerts to tilt the system from the line $y = \bar{y}$ is $m_2 g(\bar{y} - y_2)$.
- In general, the force that mass m_k exerts to tilt the system from the line $y = \bar{y}$ is $m_k g(\bar{y} - y_k)$.

For the system to balance, we need to forces to sum to 0, so that

$$\sum_{k=1}^{N} m_k g(\bar{y} - y_k) = 0 \implies \bar{y} = \frac{\sum_{k=1}^{N} m_k y_k}{\sum_{k=1}^{N} m_k}.$$

A similar argument shows that

$$\sum_{k=1}^{N} m_k g(\bar{x} - x_k) = 0 \implies \bar{x} = \frac{\sum_{k=1}^{N} m_k x_k}{\sum_{k=1}^{N} m_k}.$$

The value $M_x = \sum_{k=1}^{N} m_k y_k$ is called the **total moment** with respect to the x-axis; $M_y = \sum_{k=1}^{N} m_k x_k$ is called the **total moment** with respect to the y-axis. And the center of mass of a point-mass system is given by

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{M}, \frac{M_x}{M} \right).$$

Center of Mass - continuous system

Now, suppose that rather than a point-mass system, we have a continuous lamina with a variable mass-density $\delta(x, y)$. We may estimate its center of mass by partitioning the lamina into mn subrectangles of equal area ΔA, and treating the resulting partitioned lamina as a point-mass system. In particular, we select a point x^*_{ij}, y^*_{ij} in the ijth subrectangle, and observe that the quantity $\delta(x^*_{ij}, y^*_{ij})\Delta A$ approximates the mass of the small portion of the lamina determined by the subrectangle R_{ij}.

We now treat $\delta(x^*_{ij}, y^*_{ij})\Delta A$ as a point mass at the point (x^*_{ij}, y^*_{ij}). The coordinates $(\bar{x}, \bar{y})$ of the center of mass of these mn point masses are thus given by

$$\bar{x} = \frac{\sum_{j=1}^{n} \sum_{i=1}^{m} x_{ij}^{*} \delta(x_{ij}^{*}, y_{ij}^{*}) \Delta A}{\sum_{j=1}^{n} \sum_{i=1}^{m} \delta(x_{ij}^{*}, y_{ij}^{*}) \Delta A}, \quad \bar{y} = \frac{\sum_{j=1}^{n} \sum_{i=1}^{m} y_{ij}^{*} \delta(x_{ij}^{*}, y_{ij}^{*}) \Delta A}{\sum_{j=1}^{n} \sum_{i=1}^{m} \delta(x_{ij}^{*}, y_{ij}^{*}) \Delta A}.$$

If we take the limit as m and n go to infinity, we obtain the exact center of mass $(\bar{x}, \bar{y})$ of the continuous lamina.

The center of mass of a lamina

The coordinates $(\bar{x}, \bar{y})$ of the **center of mass of a lamina** D wuth density $\delta = \delta(x, y)$ are given by

$$\bar{x} = \frac{\iint_D x\delta(x, y) dA}{\iint \delta(x, y) dA}, \quad \bar{y} = \frac{\iint_D y\delta(x, y) dA}{\iint \delta(x, y) dA}.$$

The center of mass of a lamina can then be thought of as a **weighted average** of all of the points in the lamina with the weights given by the density at each point.

The numerators of $\bar{x}$ and $\bar{y}$ are called the respective moments of the lamina about the coordinate axes. Thus, the moment of a lamina D with density $\delta = \delta(x, y)$ about the y-axis is

$$M_y = \iint_D x\delta(x, y)dA$$

and the moment of D anout the x-axis is

$$M_x = \iint_D y\delta(x, y)dA.$$

If M is the mass of the lamina, it follows that the center of mass is

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{M}, \frac{M_x}{M}\right).$$

Example

In this example we determine integrals that represent the center of mass of a lamina D described by the triangular region bounded by the x-axis and the lines $x = 1$ and $y = 2x$ in the first quadrant if the density at point (x, y) is $\delta(x, y) = 6x + 6y + 6$.

1. Find the mass of the lamina.
2. Find the coordinates of the center of mass of the lamina.

Probability

Example

Suppose that a dart board is in the form of a disk D with radius 10 inches. If we assume that a player throws a dart at random, and is not aiming at any particular point, then it is equally probable that the dart will strike any single point on the board. The probability that the dart strikes a point in the disk D_3 of radius 3 inches is given by $\dfrac{A(D_3)}{A(D)}$. In other words, the probability can be rewritten as

$$\frac{9\pi}{100\pi} = \iint_{D_3} \frac{1}{100\pi} dA.$$

The integrand, $\frac{1}{100\pi}$ is a distribution function, that describes how the dart strikes are distributed across the board. In this case the distribution function is constant since we are assuming a uniform distribution. We can have other distribution functions like $f(x, y) = Ke^{-x^2-y^2}$ for some positive constant K. If we assume that the player is consistent enough so that the dart always strikes the board, then the probability that the dart strikes the board somewhere is 1, and the distribution function must satisfy

$$P((x, y) \in D) = \iint_D f(x, y) dA = 1.$$

Indeed, the probability that the dart strikes in any region R that lies within D is given by

$$P((x, y) \in R) = \iint_R f(x, y) dA.$$

We assume we have a **joint probability density function** f, a function of two independent variables x and y defined on a domain D that satisfies the conditions

- $f(x, y) \geq 0$ for all (x, y) in D.
- For a region R lies within D, we have

$$P((x, y) \in R) = \iint_R f(x, y)dA.$$

- The probability that the point (x, y) is in D is 1, that is

$$\iint_D f(x, y)dA = 1.$$

Note that it is possible that D could be an infinite region and the limits on the integral in equation above could be infinite.

Activity 11.4.5. A firm manufactures smoke detectors. Two components for the detectors come from different suppliers — one in Michigan and one in Ohio. The company studies these components for their reliability and their data suggests that if x is the life span (in years) of a randomly chosen component from the Michigan supplier and y the life span (in years) of a randomly chosen component from the Ohio supplier, then the joint probability density function f might be given by

$$f(x,y) = e^{-x}e^{-y}.$$

a. Theoretically, the components might last forever, so the domain D of the function f is the set D of all (x,y) such that $x \geq 0$ and $y \geq 0$. To show that f is a probability density function on D we need to demonstrate that

$$\int\int_D f(x,y)\,dA = 1,$$

or that

$$\int_0^\infty \int_0^\infty f(x,y)\,dy\,dx = 1.$$

Use your knowledge of improper integrals to verify that f is indeed a probability density function.

b. Assume that the smoke detector fails only if both of the supplied components fail. To determine the probability that a randomly selected detector will fail within one year, we will need to determine the probability that the life span of each component is between 0 and 1 years. Set up an appropriate iterated integral, and evaluate the integral to determine the probability.

c. What is the probability that a randomly chosen smoke detector will fail between years 3 and 7?

d. Suppose that the manufacturer determines that one of the components is more likely to fail than the other, and hence conjectures that the probability density function is instead $f(x, y) = Ke^{-x}e^{-2y}$. What is the value of K?

We have now studied

- curves in $\mathbb{R}^3$ can be defined parametrically by functions of the form $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$.
- surfaces can be represented by functions $z = f(x, y)$.

In this section, we will see how we can also define surfaces parametrically.

We already known that, a **one-dimensional curve** in space results from a vector function that relies upon **one parameter**, so a **two-dimensional surface** naturally involves the use of **two parameters**.

If $x = x(s,t), y = y(s,t)$, and $z = z(s,t)$ are functions of independent parameters s and t, then the terminal points of all vectors of the form

$$\mathbf{r}(s,t) = x(s,t)\mathbf{i} + y(s,t)\mathbf{j} + z(s,t)\mathbf{k}$$

form a surface in space. The equations

$$x = x(s,t), \quad y = y(s,t), \quad z = z(s,t)$$

are the parametric equations for the surface, or a parametrization of the surface.

Preview Activity 11.6.1. Recall the standard parameterization of the unit circle that is given by

$$x(t) = \cos(t) \quad \text{and} \quad y(t) = \sin(t),$$

where $0 \leq t \leq 2\pi$.

a. Determine a parameterization of the circle of radius 1 in $\mathbb{R}^3$ that has its center at $(0, 0, 1)$ and lies in the plane $z = 1$.

b. Determine a parameterization of the circle of radius 1 in 3-space that has its center at $(0, 0, -1)$ and lies in the plane $z = -1$.

c. Determine a parameterization of the circle of radius 1 in 3-space that has its center at $(0, 0, 5)$ and lies in the plane $z = 5$.

d. Taking into account your responses in (a), (b), and (c), describe the graph that results from the set of parametric equations

$$x(s,t)=\cos(t),\quad y(s,t)=\sin(t),\quad \text{and} \quad z(s,t)=s,$$

where $0 \leq t \leq 2\pi$ and $-5 \leq s \leq 5$. Explain your thinking.

e. Just as a cylinder can be viewed as a “stack” of circles of constant radius, a cone can be viewed as a stack of circles with varying radius. Modify the parametrizations of the circles above in order to construct the parameterization of a cone whose vertex lies at the origin, whose base radius is 4, and whose height is 3, where the base of the cone lies in the plane $z=3$. Use appropriate technology to plot the parametric equations you develop. (Hint: The cross sections parallel to the xy-plane are circles, with the radii varying linearly as z increases.)

Parametric Surfaces

- In single- variable setting, any function may have its graph expressed parametrically. For example, $y = g(x)$ can be parametrically described by $\langle t, g(t) \rangle$. $x^2 + y^2 = 1$ is not a function, but it can be parameterized by $\langle \cos(t), \sin(t) \rangle$, where $0 \leq t \leq 2\pi$.
- In the same way, in a two-variable setting, the surface $z = f(x, y)$ may be expressed parametrically by considering

$$\langle x(s,t), y(s,t), z(s,t) \rangle = \langle s, t, f(s,t) \rangle,$$

  where (s, t) varies over the entire domain of f. What is more powerful is that there are surfaces that cannot be generated by a single function $z = f(x, y)$ (such as the unit sphere), but that can be represented parametrically.

Example 11.6.1. Consider the torus (or doughnut) shown in Figure 11.6.2.

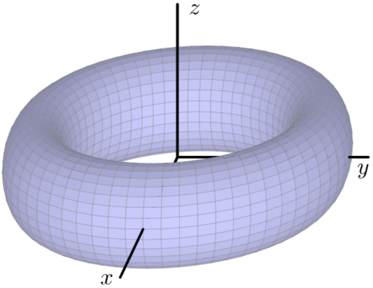


Figure 11.6.2. A torus.

To find a parametrization of this torus, recall that in Activity 11.6.1, a circle of radius r that has its center at the point $(0, 0, z_0)$ and is contained in the horizontal plane $z = z_0$ can be parametrized by

$$\mathbf{r}(t) = \langle r\cos(t), r\sin(t), z_0\rangle,$$

where $0 \leq t \leq 2\pi$.

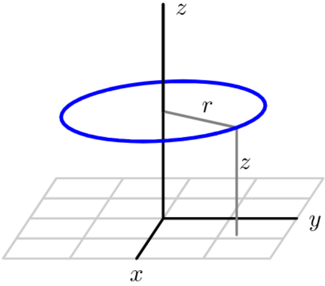


Figure 11.6.3. A circle in a horizontal plane centered at $(0, 0, z_0)$.

To obtain the torus, we begin with a circle of radius a in the xz-plane centered at $(b, 0)$, as shown on the left of Figure 11.6.4. We may parametrize the points on this circle, using the parameter s, by using the equations

$$x(s) = b + a\cos(s), \quad z(s) = a\sin(s),\ 0 \leq s \leq 2\pi.$$

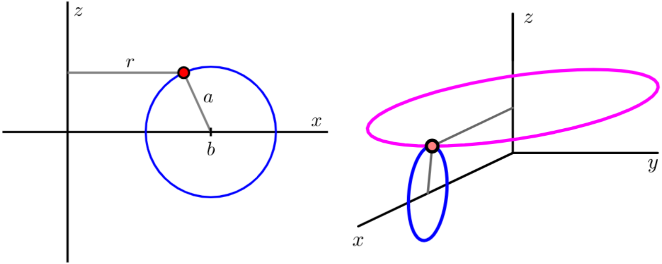


Figure 11.6.4. Revolving a circle to obtain a torus.

Let's focus our attention on one point on this circle, such as the indicated point, which has coordinates $(x(s), 0, z(s))$ for a fixed value of the parameter s (angle in the small circle). When this point is revolved about the z-axis, we obtain a circle contained in a horizontal plane centered at $0, 0, z(s)$ and having radius $x(s)$, as shown on the right of Figure 11.6.4. If we let t be the new parameter that generates the circle for the rotation about the z-axis (angle in the large circle), this circle may be parametrized by

$$\mathbf{r}(s,t) = x(s)\cos(t)\mathbf{i} + x(s)\sin(t)\mathbf{j} + z(s)\mathbf{k}.$$

Now using our earlier parametric equations for $x(s)$ and $z(s)$ for the original smaller circle, we have an overall parameterization of the torus given by

$$\mathbf{r}(s,t) = (b + a\cos(s))\cos(t)\mathbf{i} + (b + a\cos(s))\sin(t)\mathbf{j} + a\sin(s)\mathbf{k},$$

where $0 \leq s \leq 2\pi, 0 \leq t \leq 2\pi$.

Activity 11.6.2. In this activity, we seek a parametrization of the sphere of radius R centered at the origin, as shown on the left in Figure 11.6.5. Notice that this sphere may be obtained by revolving a half-circle contained in the xz-plane about the z-axis, as shown on the right.

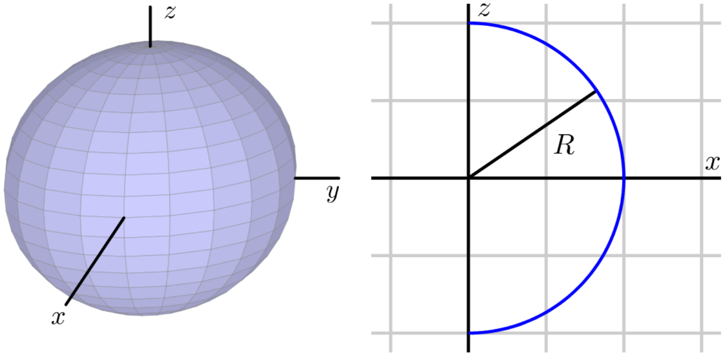


Figure 11.6.5. A sphere obtained by revolving a half-circle.

a. Begin by writing a parametrization of this half-circle using the parameter s:

$$x(s) = \dots \qquad z(s) = \dots.$$

Be sure to state the domain of the parameter s.

b. By revolving the points on this half-circle about the z-axis, obtain a parametrization $\mathbf{r}(s, t)$ of the points on the sphere of radius R. Be sure to include the domain of both parameters s and t. (Hint: What is the radius of the circle obtained when revolving a point on the half-circle around the z axis?)

c. Draw the surface defined by your parameterization with appropriate technology.

The Surface Area of Parametrically Defined Surfaces

Recall that a differentiable function is locally linear - that is, if we zoom in on the surface around a point, the surface looks like its tangent plane. To determine the surface area generated by $\langle x(s,t), y(s,t), z(s,t)\rangle$, we can use this property. Let

$$\mathbf{r}(s,t) = \langle x(s,t), y(s,t), z(s,t)\rangle$$

define a surface over a rectangular domain $a \leq s \leq b$ and $c \leq t \leq d$.

In the usual way, we slice the domain into small rectangles. In particular,

- we partition $[a, b]$ into m subintervals of length $\Delta s = \frac{b-a}{m}$ and let $a = s_0 < s_1 < \cdots < s_m = b$ be the endpoints of these subintervals.
- Also partition the interval $[c, d]$ into n subintervals of equal length $\Delta t = \frac{d-c}{n}$ and let $c = t_0 < t_1 < \cdots < t_n = d$ be the endpoints of these subintervals.

These two partitions create a partition of the rectangle $R = [a, b] \times [c, d]$ in st-coordinates into mn sub rectangles $R_{ij} = [s_{i-1}, t_{j-1}] \times [s_i, t_j]$ for $1 \le i \le m, 1 \le j \le n$. These rectangles all have equal area $\Delta s \Delta t$.

Now we want to think about the small piece of area on the surface itself that lies above one of these small rectangles in the domain. We select a small piece of domain, say, R_{ij} and try to find the area of surface above R_{ij}.

- The curve from $\mathbf{r}(s_{i-1}, t_{i-1})$ to $\mathbf{r}(s_{i-1}, t_i)$ is about $\mathbf{r}_t(s_{i-1}, t_{i-1})\Delta t$.
- The curve from $\mathbf{r}(s_{i-1}, t_{i-1})$ to $\mathbf{r}(s_i, t_{i-1})$ is about $\mathbf{r}_s(s_{i-1}, t_{i-1})\Delta s$.

So the piece of surface area above R_{ij}, denoted by S_{ij}, is about

$$\begin{aligned} S_{ij} &= |(\mathbf{r}_s(s_{i-1}, t_{i-1})\Delta s) \times (\mathbf{r}_t(s_{i-1}, t_{i-1})\Delta t)| \\ &= |\mathbf{r}_s(s_{i-1}, t_{i-1}) \times \mathbf{r}_t(s_{i-1}, t_{i-1})|\Delta s\Delta t. \end{aligned}$$

Finally, we sum the surface area approximations S_{ij} over all sub-rectangles (all i and all j) to obtain an estimate for the total surface area, S, given by

$$S \approx \sum_{i=1}^{m} \sum_{j=1}^{n} |\mathbf{r}_s(s_{i-1}, t_{i-1}) \times \mathbf{r}_t(s_{i-1}, t_{i-1})| \Delta s \Delta t.$$

Taking the limit as $m, n \to \infty$ shows that the surface area of the surface defined by $\mathbf{r}$ over the domain D is given as follows.

Surface Area

Let $\mathbf{r}(s,t) = \langle x(s,t), y(s,t), z(s,t) \rangle$ be a parameterization of a smooth surface over a domain D. The **area of the surface** defined by $\mathbf{r}$ on D is given by

$$S = \iint_D |\mathbf{r}_s \times \mathbf{r}_t| dA.$$

Activity 11.6.3. Consider the cylinder with radius a and height h defined parametrically by

$$\mathbf{r}(s,t) = a\cos(s)\mathbf{i} + a\sin(s)\mathbf{j} + t\mathbf{k}$$

for $0 \leq s \leq 2\pi$ and $0 \leq t \leq h$, as shown in Figure 11.6.7.

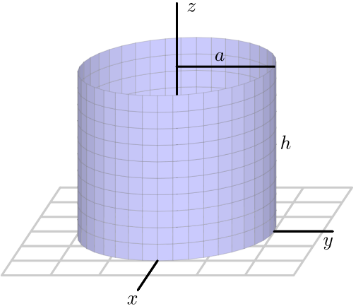


Figure 11.6.7. A cylinder.

Questions

1. Set up an iterated integral to determine the surface area of this cylinder.
2. Evaluate the iterated integral.
3. Recall that one way to think about the surface area of a cylinder is to cut the cylinder horizontally and find the perimeter of the resulting cross sectional circle, then multiply by the height. Calculate the surface area of the given cylinder using this alternate approach, and compare your work in (b).

As we noted earlier, we can take any surface $z = f(x, y)$ and generate a corresponding parameterization for the surface by writing $\langle s, t, f(s, t)\rangle$. Hence, we can use our recent work with parametrically defined surfaces to find the surface area that is generated by a function $f = f(x, y)$ over a given domain.

Activity 11.6.4. Let $z = f(x, y)$ define a smooth surface, and consider the corresponding parameterization $\mathbf{r}(s, t) = \langle s, t, f(s, t) \rangle$.

a. Let D be a region in the domain of f. Using Equation (11.6.2), show that the area, S, of the surface defined by the graph of f over D is

$$S = \iint_D \sqrt{(f_x(x, y))^2 + (f_y(x, y))^2 + 1}\, dA.$$

b. Use the formula developed in (a) to calculate the area of the surface defined by $f(x, y) = \sqrt{4 - x^2}$ over the rectangle $D = [-2, 2] \times [0, 3]$.

c. Observe that the surface of the solid describe in (b) is half of a circular cylinder. Use the standard formula for the surface area of a cylinder to calculate the surface area in a different way, and compare your result from (b).

We have now learned that we define the double integral of a continuous function $f = f(x, y)$ over a rectangle $[a, b] \times [c, d]$ as a limit of a double Riemann sum, and that these ideas parallel the single-variable integral of a function $g = g(x)$ on an interval $[a, b]$.

It is natural to wonder if it is possible to extend these ideas of double Riemann sums and double integrals for functions of two variables to triple Riemann sums and then triple integrals for functions of three variables.

Preview Activity 11.7.1. Consider a solid piece of granite in the shape of a box $B = \{(x, y, z) : 0 \leq x \leq 4, 0 \leq y \leq 6, 0 \leq z \leq 8\}$, whose density varies from point to point. Let $\delta(x, y, z)$ represent the mass density of the piece of granite at point (x, y, z) in kilograms per cubic meter (so we are measuring x, y, and z in meters). Our goal is to find the mass of this solid.

Recall that if the density was constant, we could find the mass by multiplying the density and volume; since the density varies from point to point, we will use the approach we did with two-variable lamina problems, and slice the solid into small pieces on which the density is roughly constant.

Partition the interval $[0, 4]$ into 2 subintervals of equal length, the interval $[0, 6]$ into 3 subintervals of equal length, and the interval $[0, 8]$ into 2 subintervals of equal length. This partitions the box B into sub-boxes as shown in Figure 11.7.1.

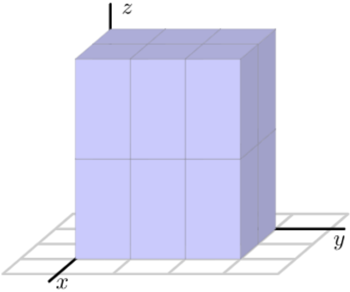


Figure 11.7.1. A partitioned three-dimensional domain.

a. Let $0 = x_0 < x_1 < x_2 = 4$ be the endpoints of the subintervals of $[0, 4]$ after partitioning. Draw a picture of Figure 11.7.1 and label these endpoints on your drawing. Do likewise with $0 = y_0 < y_1 < y_2 < y_3 = 6$ and $0 = z_0 < z_1 < z_2 = 8$ What is the length Δx of each subinterval $[x_{i-1}, x_i]$ for i from 1 to 2? the length of Δy? of Δz?

b. The partitions of the intervals $[0, 4]$, $[0, 6]$ and $[0, 8]$ partition the box B into sub-boxes. How many sub-boxes are there? What is volume ΔV of each sub-box?

c. Let B_{ijk} denote the sub-box $[x_{i-1}, x_i] \times [y_{j-1}, y_j] \times [z_{k-1}, z_k]$. Say that we choose a point $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ in the i, j, kth sub-box for each possible combination of i, j, k. What is the meaning of $\delta(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$? What physical quantity will $\delta(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\Delta V$ approximate?

d. What final step(s) would it take to determine the exact mass of the piece of granite?

Triple Riemann Sums and Triple Integrals

Definition: triple Riemann sum

Let $f = f(x, y, z)$ be a continuous function on a box $B = [a, b] \times [c, d] \times [r, s]$. The **triple Riemann sum of f over B** is created as follows.

- Partition the interval $[a, b]$ into m subintervals of equal length $\Delta x = \frac{b-a}{m}$. Let $a = x_0 < x_1 < \cdots < x_m = b$ be the endpoints of these subintervals. Do likewise with the interval $[c, d]$ using n subintervals of equal length $\Delta y = \frac{d-c}{n}$ to generate $c = y_0 < y_1 < \cdots < y_n = d$, and with the interval $[r, s]$ using ℓ subintervals of equal length $\Delta z = \frac{s-r}{\ell}$ to have $r = z_0 < z_1 < \cdots < z_\ell = s$.

Triple Riemann sum ct'd

- Let B_{ijk} be the sub-box of B with opposite vertices $(x_{i-1}, y_{j-1}, z_{k-1})$ and (x_i, y_j, z_k) for $1 \leq i \leq m, 1 \leq j \leq n, 1 \leq k \leq \ell$. The volume of each B_{ijk} is $\Delta V = \Delta x \Delta y \Delta z$.
- Let $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ be a point in box B_{ijk} for each i, j, and k. The resulting triple Riemann sum for f on B is

$$\sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=1}^{\ell} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk}) \Delta V.$$

If $f(x, y, z)$ represents the mass density of the box B, then, the triple Riemann sum approximates the total mass of the box B.

In order to find the exact mass of the box, we need to let $m, n, \ell \to \infty$; in this case, the finite sum of the mass approximations becomes the actual mass of the solid B. More generally, we have the following definition of the triple integral.

Definition: triple integral

With following notation defined as in a triple Riemann sum, the **triple integral of f over B** is

$$\iiint_B f(x,y,z)\,dV = \lim_{m,n,\ell\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=1}^{\ell} f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)\Delta V.$$

Meaning of the triple integral. Notice that the graph of $f(x, y, z)$ is in $\mathbb{R}^4$ so triple integral has no geometric meaning.

- If $f(x, y, z)$ represents the density of the solid B at each point (x, y, z), then
$$M = \iiint_B f(x, y, z)dV$$
is the mass of B.
- The average value of f over B is given by
$$f_{\mathrm{AVG}(B)} = \frac{1}{V(B)} \iiint_B f(x, y, z)dV.$$

Triple integral in general region

Each of these quantities may actually be considered over a general domain S in $\mathbb{R}^3$.

- The triple integral

$$V(S) = \iiint_S 1dV$$

  represents the volume of the solid S.

- The average value of the function $f = f(x, y, z)$ over a solid domain S is given by

$$f_{\mathrm{AVG}(S)} = \frac{1}{V(S)} \iiint_S f(x, y, z)dV.$$

- The **center of mass** of the solid S with density $\delta = \delta(x, y, z)$ is $(\bar{x}, \bar{y}, \bar{z})$, where

$$\bar{x} = \frac{\iiint_S x\delta(x, y, z)dV}{M},$$
$$\bar{y} = \frac{\iiint_S y\delta(x, y, z)dV}{M},$$
$$\bar{z} = \frac{\iiint_S z\delta(x, y, z)dV}{M},$$

and $M = \displaystyle\iiint_S \delta(x, y, z)dV$ is the mass of the solid S.

Iterated integral

In the Cartesian coordinate system, the volume element dV is $dxdydz$, and, as a consequence, a triple integral of a function f over a box $B = [a, b] \times [c, d] \times [r, s]$ in Cartesian coordinates can be evaluated as an iterated integral of the form

$$\iiint_B f(x, y, z)dV = \int_a^b \int_c^d \int_r^s f(x, y, z)dzdydx.$$

If we want to evaluate a triple integral as an iterated integral over a solid S that is not a box, then we need to describe the solid in terms of variable limits.

Activity 11.7.2.

a. Set up and evaluate the triple integral of $f(x,y,z) = x - y + 2z$ over the box $B = [-2,3] \times [1,4] \times [0,2]$.

b. Let S be the solid cone bounded by $z = \sqrt{x^2+y^2}$ and $z = 3$. A picture of S is shown at right in Figure 11.7.4. Our goal in what follows is to set up an iterated integral of the form

$$\int_{x=?}^{x=?} \int_{y=?}^{y=?} \int_{z=?}^{z=?} \delta(x,y,z)\, dz\, dy\, dx \tag{11.7.1}$$

to represent the mass of S in the setting where $\delta(x,y,z)$ tells us the density of S at the point (x,y,z). Our particular task is to find the limits on each of the three integrals.

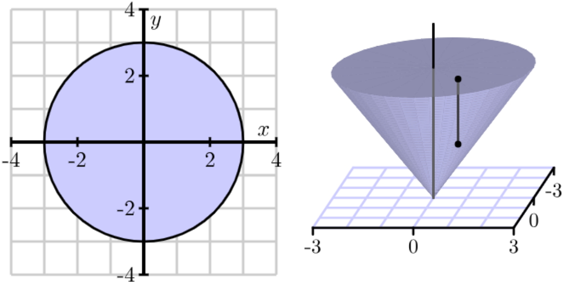


Figure 11.7.4. Left: The cone. Right: Its projection.

i. If we think about slicing up the solid, we can consider slicing the domain of the solid's projection onto the xy-plane (just as we would slice a two-dimensional region in $\mathbb{R}^2$), and then slice in the z -direction as well. The projection of the solid onto the xy-plane is shown at left in Figure 11.7.4. If we decide to first slice the domain of the solid's projection perpendicular to the x-axis, over what range of constant x-values would we have to slice?

ii. If we continue with slicing the domain, what are the limits on y on a typical slice? How do these depend on x? What, therefore, are the limits on the middle integral?

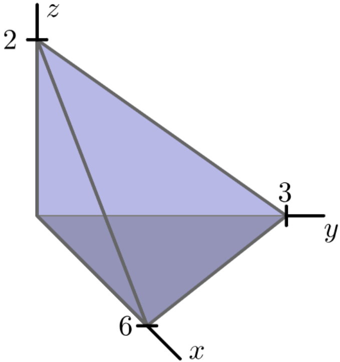
iii. Finally, now that we have thought about slicing up the two-dimensional domain that is the projection of the cone, what are the limits on z in the innermost integral? Note that over any point (x, y) in the plane, a vertical slice in the z direction will involve a range of values from the cone itself to its flat top. In particular, observe that at least one of these limits is not constant but depends on x and y.

iv. In conclusion, write an iterated integral of the form (11.7.1) that represents the mass of the cone S.

Note: When setting up iterated integrals, the limits on a given variable can be only in terms of the remaining variables. For example, we can choose 'first z and then xy' or 'first y and then xz'.

- $$\iiint_S f(x,y,z)dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{h_1(x,y)}^{h_2(x,y)} f(x,y,z)dzdydx.$$
- $$\iiint_S f(x,y,z)dV = \int_r^s \int_{p_1(z)}^{p_2(z)} \int_{q_1(x,z)}^{q_2(x,z)} f(x,y,z)dydxdz.$$

Example 11.7.5. Find the mass of the tetrahedron in the first octant bounded by the coordinate planes and the plane $x + 2y + 3z = 6$ if the density at point (x, y, z) is given by $\delta(x, y, z) = x + y + z$. A picture of the solid tetrahedron is shown at left in Figure 11.7.6.



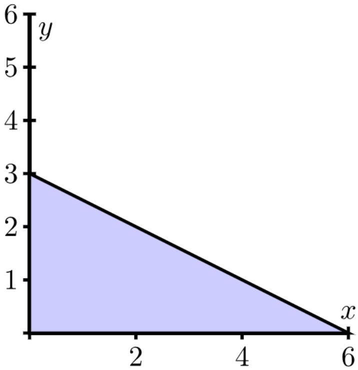


Figure 11.7.6. Left: The tetrahedron. Right: Its projection.

Question

What is the mass, M, of the tetrahedron?

Solution

The mass can be found by the following triple integral

$$M = \iint_S \delta(x, y, z) dV,$$

where S is the solid tetrahedron described above. In this example, we have several ways to solve the triple integral.

Solution 1

We choose to integrate w.r.t. z first for the innermost integral. We can determine the bounds for x first and then yz. The range of x is just from 0 to 6. For each x, we have

$$2y + 3z = 6 - x.$$

The cross section cut by the plane parallel to yz plane is

$$\begin{cases} 0 \le y \le \frac{1}{2}(6-x) \\ z \ge 0 \\ 2y + 3z \le 6 - x \end{cases} \implies \begin{cases} 0 \le y \le \frac{1}{2}(6-x) \\ 0 \le z \le \frac{1}{3}(6-x-2y) \end{cases}$$

Solution 1 ct'd

We can also determine the bounds for z first and then xy. The top of the tetrahedron is given by the equation

$$x + 2y + 3z = 6;$$

solving for z then yields

$$z = \frac{1}{3}(6 - x - 2y).$$

The bottom of the tetrahedron is the xy-plane, so the limits on z in the iterated integral will be $0 \leq z \leq \frac{1}{3}(6 - x - 2y)$. Then, to find the bounds on x and y we project the tetrahedron onto the xy-plane; this corresponds to setting $z = 0$ in the equation $z = \frac{1}{3}(6 - x - 2y)$. The resulting relation between x and y is $x + 2y = 6$.
If we choose to integrate with respect to y for the middle integral in the iterated integral, we can get the bounds on y relied on x. It is $0 \leq y \leq \frac{1}{2}(6 - x)$. And finally the bounds on x is from 0 to 6.

Solution 1 ct'd

Therefore, the triple integral can be transformed into the following iterated integral:

$$
\begin{aligned}
&\int_0^6 \int_0^{\frac{1}{2}(6-x)} \int_0^{\frac{1}{3}(6-x-2y)} (x+y+z)dzdydx \\
&= \int_0^6 \int_0^{\frac{1}{2}(6-x)} \left[xz + yz + \frac{z^2}{2}\right]\Big|_0^{\frac{1}{3}(6-x-2y)} dydx \\
&= \int_0^6 \int_0^{\frac{1}{2}(6-x)} \left[2 + \frac{4}{3}x + \frac{2}{3}y - \frac{5}{18}x^2 - \frac{4}{9}y^2 - \frac{7}{9}xy\right] dydx \\
&= \int_0^6 \left[2y + \frac{4}{3}xy + \frac{y^2}{3} - \frac{5}{18}x^2y - \frac{7}{18}xy^2 - \frac{4}{27}y^3\right]\Big|_0^{\frac{1}{2}(6-x)} dx \\
&= \int_0^6 \left[\frac{13}{216}x^3 - \frac{7}{12}x^2 + \frac{x}{2} + 5\right] dx \\
&= \int_0^6 \left[\frac{13}{864}x^4 - \frac{7}{36}x^3 + \frac{x^2}{4} + 5x\right]\Big|_0^6 dx = \frac{33}{2}.
\end{aligned}
$$

Solution 2

We choose to integrate w.r.t. y first for the innermost integral. We can determine the bounds for z first and then xy. The range of z is just from 0 to 2. For each z, we have

$$x + 2y = 6 - 3z.$$

The cross section cut by the plane parallel to xy plane is

$$\begin{cases} 0 \leq x \leq 6 - 3z \\ y \geq 0 \\ x + 2y \leq 6 - 3z \end{cases} \Longrightarrow \begin{cases} 0 \leq x \leq 6 - 3z \\ 0 \leq y \leq \frac{1}{2}(6 - x - 3z) \end{cases}$$

Therefore, the triple integral can be transformed into the following iterated integral:

$$\int_0^2 \int_0^{6-3z} \int_0^{\frac{1}{2}(6-x-3z)} (x + y + z) dy dx dz.$$

There are several other ways we could have set up the integral to give the mass of the tetrahedron in the last example.

Question

- Set up an iterated integral, integrating first with respect to z, then x, then y that is equivalent to the integral

$$\int_0^6 \int_0^{\frac{1}{2}(6-x)} \int_0^{\frac{1}{3}(6-x-2y)} (x+y+z)dzdydx.$$

- Set up an iterated integral, integrating first with respect to y, then z, then x that is equivalent to the integral above.
- Set up an iterated integral, integrating first with respect to x, then y, then z that is equivalent to the integral above.

Activity 11.7.4. A solid S is bounded below by the square $z = 0$, $-1 \leq x \leq 1$, $-1 \leq y \leq 1$ and above by the surface $z = 2 - x^2 - y^2$. A picture of the solid is shown in Figure 11.7.7.

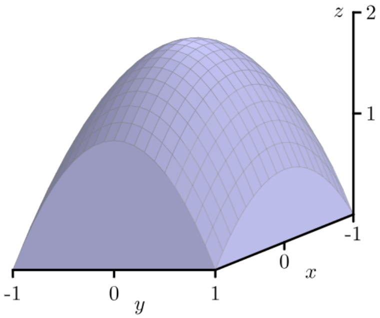


Figure 11.7.7. The solid bounded by the surface $z = 2 - x^2 - y^2$.

a. First, set up an iterated double integral to find the volume of the solid S as a double integral of a solid under a surface. Then set up an iterated triple integral that gives the volume of the solid S. You do not need to evaluate either integral. Compare the two approaches.

b. Set up (but do not evaluate) iterated integral expressions that will tell us the center of mass of S, if the density at point (x, y, z) is $\delta(x, y, z) = x^2 + 1$.

c. Set up (but do not evaluate) an iterated integral to find the average density on S using the density function from part (b).

d. Use technology appropriately to evaluate the iterated integrals you determined in (a), (b), and (c); does the location you determined for the center of mass make sense?

§11.8 Triple Integrals in Cylindrical and Spherical Coordinates

We have encountered two different coordinate systems in $\mathbb{R}^2$ - the rectangular and polar coordinates systems. In a similar way, there are two additional natural coordinate systems in $\mathbb{R}^3$. Given that we are already familiar with the Cartesian coordinate system for $\mathbb{R}^3$, we next investigate the cylindrical and spherical coordinate systems.

Preview Activity 11.8.1. In the following questions, we investigate the two new coordinate systems that are the subject of this section: cylindrical and spherical coordinates. Our goal is to consider some examples of how to convert from rectangular coordinates to each of these systems, and vice versa. Triangles and trigonometry prove to be particularly important.

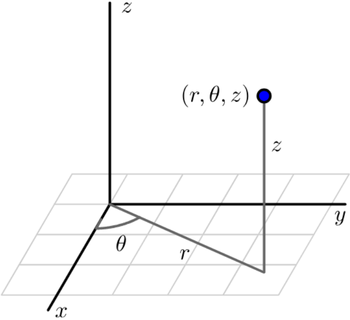


Figure 11.8.1. The cylindrical (left) and spherical (right) coordinates of a point.

a. The cylindrical coordinates of a point in $\mathbb{R}^3$ are given by (r, θ, z) where r and θ are the polar coordinates of the point (x, y) and z is the same z coordinate as in Cartesian coordinates. An illustration is given at left in Figure 11.8.1.

   i. Find cylindrical coordinates for the point whose Cartesian coordinates are $(-1, \sqrt{3}, 3)$. Draw a labeled picture illustrating all of the coordinates.

   ii. Find the Cartesian coordinates of the point whose cylindrical coordinates are $\left(2, \frac{5\pi}{4}, 1\right)$. Draw a labeled picture illustrating all of the coordinates.

b. The spherical coordinates of a point in $\mathbb{R}^3$ are ρ (rho), θ, and ϕ (phi), where ρ is the distance from the point to the origin, θ has the same interpretation it does in polar coordinates, and ϕ is the angle between the positive z axis and the vector from the origin to the point, as illustrated at right in Figure 11.8.1. You should convince yourself that any point in $\mathbb{R}^3$ can be represented in spherical coordinates with $\rho \geq 0$, $0 \leq \theta < 2\pi$, and $0 \leq \phi \leq \pi$.

For the following questions, consider the point P whose Cartesian coordinates are $(-2, 2, \sqrt{8})$.

i. What is the distance from P to the origin? Your result is the value of ρ in the spherical coordinates of P.

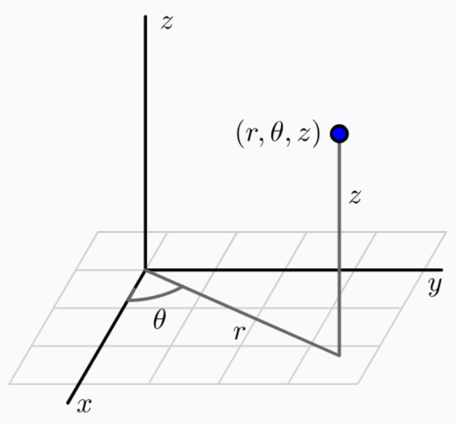
ii. Determine the point that is the projection of P onto the xy-plane. Then, use this projection to find the value of θ in the polar coordinates of the projection of P that lies in the plane. Your result is also the value of θ for the spherical coordinates of the point.

iii. Based on the illustration in Figure 11.8.1, how is the angle ϕ determined by ρ and the z coordinate of P? Use a well-chosen right triangle to find the value of ϕ, which is the final component in the spherical coordinates of P. Draw a carefully labeled picture that clearly illustrates the values of ρ, θ, and ϕ in this example, along with the original rectangular coordinates of P.

iv. Based on your responses to i., ii., and iii., if we are given the Cartesian coordinates (x, y, z) of a point Q, how are the values of ρ, θ, and ϕ in the spherical coordinates of Q determined by x, y, and z?

Cylindrical Coordinates

As we stated in Preview Activity 11.8.1, the cylindrical coordinates of a point are (r, θ, z), where r and θ are the polar coordinates of the point (x, y), and z is the same z coordinate as in Cartesian coordinates.



Converting between Cartesian and cylindrical coordinates.

- *Converting from Cartesian to cylindrical..*

  If we are given the Cartesian coordinates (x, y, z) of a point P, then the cylindrical coordinates (r, θ, z) of P satisfy

  $$x = r\cos(\theta) \quad y = r\sin(\theta) \quad \text{and} \quad z = z.$$

- *Converting from cylindrical to cartesian..*

  If we are given the cylindrical coordinates (r, θ, z) of a point P, then the Cartesian coordinates (x, y, z) of P satisfy

  $$r^2 = x^2 + y^2 \quad \tan(\theta) = \frac{y}{x} \quad \text{and} \quad z = z,$$

  assuming $x \neq 0$.

Activity 11.8.2. In this activity, we graph some surfaces using cylindrical coordinates. To improve your intuition and test your understanding, you should first think about what each graph should look like before you plot it using appropriate technology.

a. What familiar surface is described by the points in cylindrical coordinates with $r = 2$, $0 \leq \theta \leq 2\pi$, and $0 \leq z \leq 2$? How does this example suggest that we call these coordinates *cylindrical coordinates*? How does your answer change if we restrict θ to $0 \leq \theta \leq \pi$?

b. What familiar surface is described by the points in cylindrical coordinates with $\theta = 2$, $0 \leq r \leq 2$, and $0 \leq z \leq 2$?

c. What familiar surface is described by the points in cylindrical coordinates with $z = 2$, $0 \leq \theta \leq 2\pi$, and $0 \leq r \leq 2$?

d. Plot the graph of the cylindrical equation $z = r$, where $0 \leq \theta \leq 2\pi$ and $0 \leq r \leq 2$. What familiar surface results?

e. Plot the graph of the cylindrical equation $z = \theta$ for $0 \leq \theta \leq 4\pi$. What does this surface look like?

Triple Integrals in Cylindrical Coordinates

To evaluate a triple integral $\iiint_S f(x, y, z) dV$ as an iterated integral in Cartesian coordinates, we use the fact that the volume element $dV = dzdydx$ (which corresponds to the volume of a small box). To evaluate a triple integral in cylindrical coordinates, we similarly must understand the volume element dV in cylindrical coordinates.

Activity 11.8.3. A picture of a cylindrical box, $B = \{(r, \theta, z) : r_1 \leq r \leq r_2, \theta_1 \leq \theta \leq \theta_2, z_1 \leq z \leq z_2\}$, is shown in Figure 11.8.2. Let $\Delta r = r_2 - r_1$, $\Delta\theta = \theta_2 - \theta_1$, and $\Delta z = z_2 - z_1$. We want to determine the volume ΔV of B in terms of Δr, $\Delta\theta$, Δz, r, θ, and z.

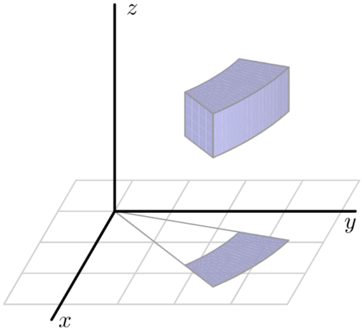


Figure 11.8.2. A cylindrical box.

Questions

a. Appropriately label $\Delta r, \Delta\theta$ and Δz in Figure 11.8.2.

b. Let ΔA be the area of the projection of the box, B, onto the xy-plane, which is shaded blue in Figure 11.8.2. Recall that we previously determined the area ΔA in polar coordinates in terms of r, Δr, and $\Delta\theta$. In light of the fact that we know ΔA and that z is the standard z coordinate from Cartesian coordinates, what is the volume ΔV in cylindrical coordinates?

Since dV in cylindrical coordinates is given by $dV = rdzdrd\theta$, we have the following rules.

Triple integrals in cylindrical coordinates

Given a continuous function $f = f(x, y, z)$ over a region S in $\mathbb{R}^3$,

$$\iiint_S f(x, y, z)dV = \iiint_S f(r\cos(\theta), r\sin(\theta), z)rdzdrd\theta.$$

Of course, to complete the task of writing an iterated integral in cylindrical coordinates, we need to determine the limits on the three integrals: r, θ and z.

Activity 11.8.4. In this activity we work with triple integrals in cylindrical coordinates.

a. Let S be the solid bounded above by the graph of $z = x^2 + y^2$ and below by $z = 0$ on the unit disk in the xy-plane.

   i. The projection of the solid S onto the xy-plane is a disk. Describe this disk using polar coordinates.

   ii. Now describe the surfaces bounding the solid S using cylindrical coordinates.

   iii. Determine an iterated triple integral expression in cylindrical coordinates that gives the volume of S. You do not need to evaluate this integral.

b. Suppose the density of the cone defined by $r = 1 - z$, with $z \geq 0$, is given by $\delta(r, \theta, z) = z$. A picture of the cone is shown at left in Figure 11.8.3, and the projection of the cone onto the xy-plane in given at right in Figure 11.8.3. Set up an iterated integral in cylindrical coordinates that gives the mass of the cone. You do not need to evaluate this integral.

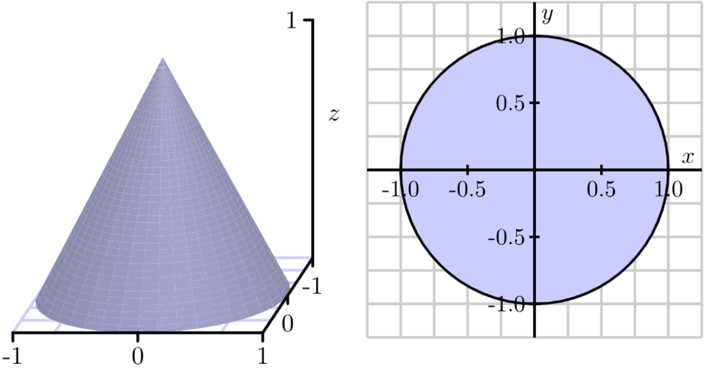


Figure 11.8.3. The cylindrical cone $r = 1 - z$ and its projection onto the xy-plane.

c. Determine an iterated integral expression in cylindrical coordinates whose value is the volume of the solid bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the cone $z = 4 - \sqrt{x^2 + y^2}$. A picture is shown in Figure 11.8.4. You do not need to evaluate this integral.

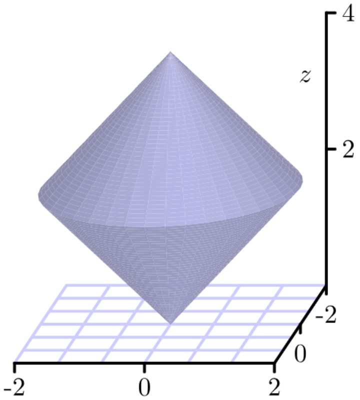
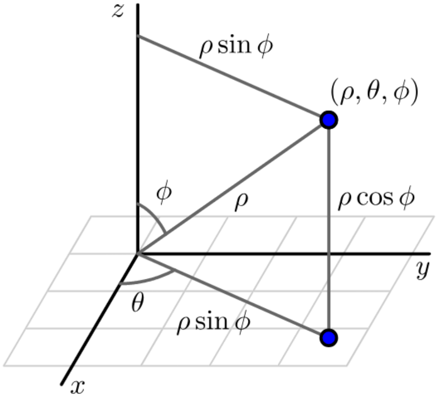


Figure 11.8.4. A solid bounded by the cones $z = \sqrt{x^2 + y^2}$ and $z = 4 - \sqrt{x^2 + y^2}$.

Spherical Coordinates

As we saw in Preview Activity 11.8.1, the spherical coordinates of a point in 3-space have the form (ρ, θ, ϕ), where ρ is the distance from the point to the origin, θ has the same meaning as in polar coordinates, and ϕ is the angle between the positive z axis and the vector from the origin to the point.



Coverting between Cartesian and spherical coordinates.

- *Converting from Cartesian to spherical..*

  If we are given the Cartesian coordinates (x, y, z) of a point P, then the spherical coordinates (ρ, θ, ϕ) of P satisfy

  $$\rho = \sqrt{x^2 + y^2 + z^2} \qquad \tan(\theta) = \frac{y}{x} \qquad \text{and} \qquad \cos(\phi) = \frac{z}{\rho},$$

  where in the latter two equations, we require $x \neq 0$ and $\rho \neq 0$.

- *Converting from spherical to Cartesian..*

  If we are given the spherical coordinates (ρ, θ, ϕ) of a point P, then the Cartesian coordinates (x, y, z) of P satisfy

  $$x = \rho \sin(\phi) \cos(\theta) \qquad y = \rho \sin(\phi) \sin(\theta) \qquad \text{and} \qquad z = \rho \cos(\phi).$$

Like in polar coordinates, in spherical coordinates, we often view ρ as a function of θ and ϕ, thus viewing distance from the origin as a function of two key angles.

Activity 11.8.5. In this activity, we graph some surfaces using spherical coordinates. To improve your intuition and test your understanding, you should first think about what each graph should look like before you plot it using appropriate technology.

a. What familiar surface is described by the points in spherical coordinates with $\rho = 1$, $0 \leq \theta \leq 2\pi$, and $0 \leq \phi \leq \pi$? How does this particular example demonstrate the reason for the name of this coordinate system? What if we restrict ϕ to $0 \leq \phi \leq \frac{\pi}{2}$?

b. What familiar surface is described by the points in spherical coordinates with $\phi = \frac{\pi}{3}$, $0 \leq \rho \leq 1$, and $0 \leq \theta \leq 2\pi$?

c. What familiar shape is described by the points in spherical coordinates with $\theta = \frac{\pi}{6}$, $0 \leq \rho \leq 1$, and $0 \leq \phi \leq \pi$?

d. Plot the graph of $\rho = \theta$, for $0 \leq \phi \leq \pi$ and $0 \leq \theta \leq 2\pi$. How does the resulting surface appear?

Triple Integrals in Spherical Coordinates

A triple integral $\iiint_S f(x,y,z)dV$ n spherical coordinates can be evaluated as an iterated integral once we understand the volume element dV.

Activity 11.8.6. To find the volume element dV in spherical coordinates, we need to understand how to determine the volume of a spherical box of the form $\rho_1 \leq \rho \leq \rho_2$ (with $\Delta\rho = \rho_2 - \rho_1$), $\phi_1 \leq \phi \leq \phi_2$ (with $\Delta\phi = \phi_2 - \phi_1$), and $\theta_1 \leq \theta \leq \theta_2$ (with $\Delta\theta = \theta_2 - \theta_1$). An illustration of such a box is given at left in Figure 11.8.6. This spherical box is a bit more complicated than the cylindrical box we encountered earlier. In this situation, it is easier to approximate the volume ΔV than to compute it directly. Here we can approximate the volume ΔV of this spherical box with the volume of a Cartesian box whose sides have the lengths of the sides of this spherical box. In other words,

$$\Delta V \approx |PS|\,|\widehat{PR}|\,|\widehat{PQ}|,$$

where $|\widehat{PR}|$ denotes the length of the circular arc from P to R.

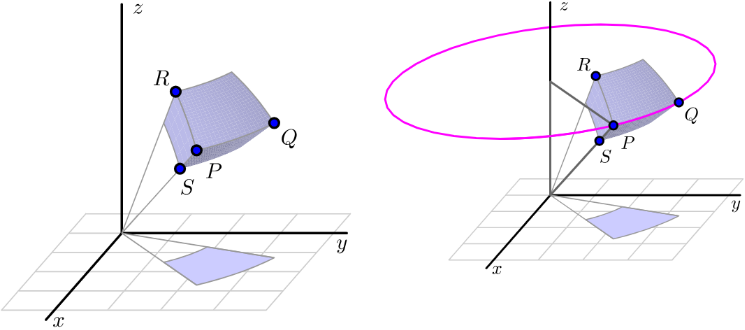


Figure 11.8.6. Left: A spherical box. Right: A spherical volume element.

a. What is the length $|PS|$ in terms of ρ?

b. What is the length of the arc $\overset{\frown}{PR}$? (Hint: The arc $\overset{\frown}{PR}$ is an arc of a circle of radius ρ_2, and arc length along a circle is the product of the angle measure (in radians) and the circle's radius.)

c. What is the length of the arc $\overset{\frown}{PQ}$? (Hint: The arc $\overset{\frown}{PQ}$ lies on a horizontal circle as illustrated at right in Figure 11.8.6. What is the radius of this circle?)

d. Use your work in (a), (b), and (c) to determine an approximation for ΔV in spherical coordinates.

Letting $\Delta\rho, \Delta\theta$, and $\Delta\phi$ go to 0, from the result in Activity 11.8.6 that $dV = \rho^2 \sin(\phi) d\rho d\theta d\phi$ in spherical coordinates, and thus allows us to state the following general rule.

Triple integrals in cylindrical coordinates

Given a continuous function $f = f(x, y, z)$ over a region S in $\mathbb{R}^3$,

$$\iiint_S f(x, y, z) dV$$
$$= \iiint_S f(\rho\sin(\phi)\cos(\theta), \rho\sin(\phi)\sin(\theta), \rho\cos(\phi))\rho^2\sin(\phi) d\rho d\theta d\phi.$$

in spherical coordinates.

Activity 11.8.7. We can use spherical coordinates to help us more easily understand some natural geometric objects.

a. Recall that the sphere of radius a has spherical equation $\rho = a$. Set up and evaluate an iterated integral in spherical coordinates to determine the volume of a sphere of radius a.

b. Set up, but do not evaluate, an iterated integral expression in spherical coordinates whose value is the mass of the solid obtained by removing the cone $\phi = \frac{\pi}{4}$ from the sphere $\rho = 2$ if the density δ at the point (x, y, z) is $\delta(x, y, z) = \sqrt{x^2 + y^2 + z^2}$. An illustration of the solid is shown in Figure 11.8.7.

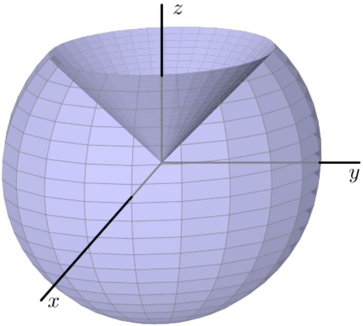


Figure 11.8.7. The solid cut from the sphere $\rho = 2$ by the cone $\phi = \frac{\pi}{4}$.